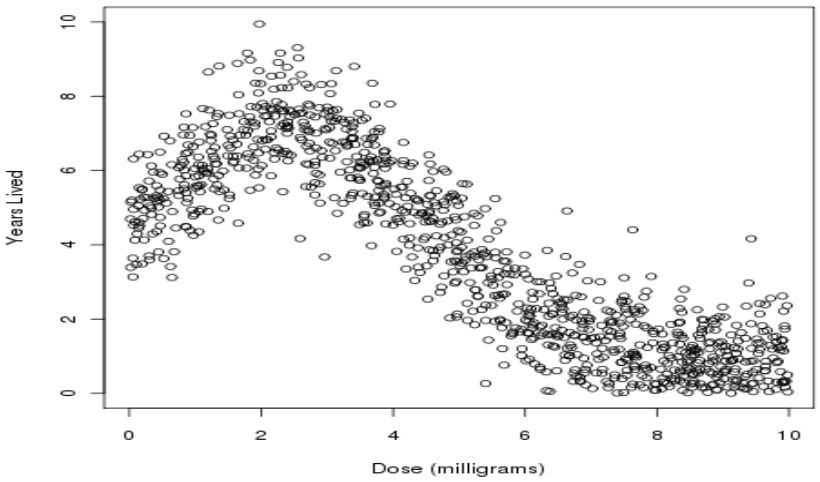
Review Questions for Lecture 7

1. If the data are football shaped, the RMS error approximates the SD inside each vertical strip. Strictly speaking, however, we do not need the data to be football shaped. All that is required is that _____ follows a straight line and that the _____ in every vertical strip is the same. Fill in the blanks and make a sketch a dataset that satisfies both of the two criteria, but **isn't** football shaped.
2. A dataset follows a linear trend and is homoscedastic, but isn't football shaped. The mean of x is 100, the mean of y is 200, and the SD of each is 20. The correlation is 0.8. If we look in a vertical strip where $x \approx 140$,
   a. What is the mean of the y values? Or is it not possible to say?
   b. What is the SD of the y values? Or is it not possible to say?
   c. What fraction of the y values are outside the range 200 to 264? Or is it not possible to say?
   d. What is the largest possible fraction of y values that are outside the range 200 to 264? Or is it not possible to say?
3. (Hypothetical) A drug company is testing a new drug meant to lengthen the lives of cancer patients. However, the company doesn't know what dose of the drug is the best to give to people. Therefore, the company decides it would be best to give each patient a random dose, ranging from 0 milligrams to 10 milligrams. Neither the patients nor their doctors will know what doses the patients get. The company finds 1,000 cancer patients who are willing to participate in this experiment. After 10 years, all of the patients have died, and the company can make a scatterplot of how many years they survived after receiving the drug. The results are shown below:

	Dose	Years Lived
Avg:	5.06	3.77
SD:	2.89	2.51

correlation: $r = -0.84$



a. True or False – This is a randomized controlled double-blind experiment. What is the control group if you believe that there is a control group?
b. True or False and explain – As the dose increases, the years lived tends to decrease as represented by the correlation coefficient.
c. True or False and explain – The drug works.
d. True or False and explain – Participating in the experiment caused many people to die early.
e. True or False and explain – To estimate the average years lived among all people who got a dose of 3 milligrams, we would want to calculate

$$\frac{3 - 5.06}{2.89} \times (-0.84) \times 2.51 + 3.77$$

f. True or False and explain – To estimate the SD of years lived among all people who got a dose of 3 milligrams, we would want to calculate

$$\sqrt{1 - 0.84^2} \times \mathrm{SD}(y)$$

4. Write an R function called MAXABS that will compute the maximum of the absolute values of a given list of numbers. Test your function on the list of numbers 2, 9, 5, -4, 0. You will want to write a loop as in last week's lab.